

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 2

Roll no: 14 PRN No: 12414973 Name of Student: Yugandhar Paulbudhe

Subject Name: Deep Learning

Problem Statement:

Design and implement a Multilayer Perceptron (MLP) for classification of the Iris or Wine dataset, and evaluate its performance using accuracy and a confusion matrix.

Algorithm:

- 1) Start the program.
- 2) Import the required libraries:
 - i. Numpy
 - ii. Matplotlib
 - iii. Pandas
 - iv. Seaborn
 - v. Scikit-learn
 - vi. TensorFlow/Keras
- 3) Load the Iris dataset (Iris.csv) into a Pandas DataFrame. Separate the dataset into:
 - i. Training data
 - ii. Testing data
- 4) Display the dataset information such as:
 - i. Shape of the dataset
 - ii. Column names
 - iii. First five records

5) Preprocess the dataset:

- i. Remove the Id column (if present).
- ii. Separate the input features (X) and target labels (y).
- iii. Convert categorical target labels into numerical values using LabelEncoder.

6) Split the dataset into training and testing sets using an 80:20 ratio.

7) Normalize the input features using StandardScaler.

8) Train the MLP model using the training dataset for 50 epochs with a batch size of 8.

9) Evaluate the trained model using the testing dataset.

10) Visualize the Confusion Matrix using a heatmap.

Code:

[25]
✓ 0s

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

[2]
✓ 0s

```
df = pd.read_csv("Iris.csv")

df.head()
```

[3]
✓ 0s

```
print("Dataset Shape:", df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nDataset Information:")
df.info()

print("\nFirst Five Rows:")
print(df.head())
```

[4]
✓ 0s

```
print(df.isnull().sum())
```

[12]
✓ 0s

```
X = df.iloc[:, 1:5]
y = df.iloc[:, 5]

print(X.head())
print(y.head())
```

Output:

```
Id          0
SepalLengthCm  0
SepalWidthCm  0
PetalLengthCm  0
PetalWidthCm  0
Species      0
dtype: int64
```

```
      SepallengthCm  SepalWidthCm  PetallengthCm  PetalWidthCm
0           5.1           3.5           1.4           0.2
1           4.9           3.0           1.4           0.2
2           4.7           3.2           1.3           0.2
3           4.6           3.1           1.5           0.2
4           5.0           3.6           1.4           0.2
0      Iris-setosa
1      Iris-setosa
2      Iris-setosa
3      Iris-setosa
4      Iris-setosa
Name: Species, dtype: object
```

```
Classes:
['Iris-setosa' 'Iris-versicolor' 'Iris-virginica']
```

```
Feature Shape: (150, 4)
Target Shape: (150,)
Feature Columns:
Index(['SepallengthCm', 'SepalWidthCm', 'PetallengthCm', 'PetalWidthCm'], dtype='object')
```

```
Training Samples: 120
Testing Samples: 30
```

```
Accuracy Score: 1.0
```

Classification Report:				
	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
Confusion Matrix:

[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```



